Project Report

Heart Disease Analysis

Team Members

Team ID: LTVIP2026TMIDS88469

Team Leader	Shaik Mohammad Asif
Member 1	Manasa G
Member 2	Mavilla Nagamma
Member 3	Mohan S

1. INTRODUCTION

1.1 Project Overview

The **Heart Disease Analysis** project is an interactive data analytics and visualization solution designed to help users understand the key **risk factors** associated with heart disease and support data-driven healthcare insights. Using **MySQL, Tableau, and Flask**, the project transforms raw patient clinical data — such as **age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, and ST depression** — into clear and meaningful insights through **interactive dashboards and a story**.

The project focuses on identifying patterns and relationships between health indicators and heart disease occurrence. By visualizing these factors, the system makes it easier to interpret medical data, compare patient groups, and highlight high-risk conditions in an understandable way.

1.2 Purpose

The main purpose of this project is to provide healthcare analysts, students, and general users with **visual and analytical insights** that reveal patterns in heart disease occurrence, risk factor distribution, and patient segmentation. This helps in understanding how different clinical parameters influence heart disease risk, supports better awareness, and improves decision-making through data.

The project also demonstrates a complete analytics pipeline by combining:

- **MySQL** for structured data storage and SQL-based analysis
- **Tableau** for interactive dashboards and story visualization
- **Flask** for web integration and online access through deployment

This ensures the analysis is not only accurate but also easily accessible and presentable for real-world use and internship evaluation.

2. IDEATION PHASE

2.1 Problem Statement

Patient / User Problem Statement :

I am...

A person who wants to understand my heart health and reduce the risk of heart disease by learning about the medical factors that influence it — such as age, blood pressure, cholesterol, chest pain symptoms, and lifestyle-related indicators..

I'm trying to...

Analyze heart disease-related health data to identify the major risk factors, understand patterns among patients, and gain clear insights that can support awareness and early risk identification.

But...

I often face difficulty in understanding which factors are truly important, how they are connected, and how to interpret complex medical data without clear visualization or structured analysis.

Because...

Heart disease datasets contain many clinical parameters, and raw data alone is hard to interpret. Also, medical insights are often presented in technical formats without interactive dashboards or easy comparisons.

Which makes me...

Feel confused, uncertain, and unable to confidently understand heart disease risk trends — leading to poor awareness, delayed prevention, and difficulty in making informed health-related decisions.

I am	A person who wants to understand my heart health and reducisk of heart disease by learning about about medical factors that influence it — such age, pressure, cholestorol, chest pain symptoms and lifestyle-relatied indicators
I'm trying to	Analyze heart disease-related lested health data to identify the the major risk factors s the factors, and inderstod pattens, anttarng patients, and gain ; and can suppovainess, and early risk identifiation
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which makes me feel	confused, uncertain, and unable to confidently understand heart disease risk trends — how to trends – leading poor awareneed prevention and diffiluty in makied o making informed healthaaith-relatied decisions

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a middle-aged person with high blood pressure	understand if my blood pressure increases my heart disease risk	I can't clearly know how serious my risk is	medical data is complex and not explained in a simple way	worried and uncertain about my health
PS-2	a person with high cholesterol levels	find out whether cholesterol strongly affects heart disease	I don't know the relationship between cholesterol and heart disease	raw reports don't show comparisons or patterns clearly	confused and anxious about future risk
PS-3	a person who experiences chest pain sometimes	identify whether my chest pain type is linked to heart disease	I can't differentiate normal pain vs risky pain	symptoms vary and there is no easy data-based explanation	scared and stressed while making decisions

2.2 Empathy Map Canvas

WHO are we empathizing with?

Primary Users:

- Doctors / Physicians (Cardiologists, General Physicians)
- Hospital administrators
- Medical researchers

- Healthcare data analysts
- Public health organizations
- Patients who want to understand their heart risk

Key Stakeholders:

- Data analysts using Tableau
- Hospital decision-makers
- Healthcare management teams
- Medical staff (nurses, clinical assistants)
- Policy makers (public health)

What do they NEED TO DO?

- Identify patients who are at high risk of heart disease early.
- Understand the key risk factors (age, cholesterol, BP, smoking, diabetes, etc.).
- Visualize complex medical data in an easy and interactive way.
- Analyze patient patterns by gender, age group, and health conditions.
- Support better clinical decisions using data insights.
- Reduce late diagnosis and prevent severe heart-related incidents.

What do they SEE?

- Large amounts of patient medical data stored in different formats.
- Hospital records that are messy or incomplete.
- Many patients with multiple risk factors at the same time.
- Increasing heart disease cases due to lifestyle changes.
- Lack of clear dashboards for quick understanding.
- Need for better prediction and prevention strategies.

What do they SAY?

- *"We need a clear dashboard to identify high-risk patients."*
- *"It's hard to analyze patient data quickly."*
- *"We need to know which factors contribute most to heart disease."*
- *"We must justify decisions using real data."*
- *"We want visual insights, not just raw medical records."*

What do they DO?

- Collect patient records from hospitals or healthcare systems.
- Store medical data in databases (MySQL).
- Use Excel or manual methods to review patient history.
- Prepare reports for doctors, hospitals, or management.
- Spend time cleaning and filtering medical datasets.
- Use experience-based judgement when insights are unclear.

What do they HEAR?

- From hospital leadership: *"We need data-driven healthcare improvements."*
- From medical experts: *"Early detection saves lives."*
- From peers: *"Dashboards make analysis faster and clearer."*
- From patients: *"We want better prevention and clear guidance."*
- From public health sector: *"Heart disease is one of the top causes of death."*

PAINS

- Patient data is inconsistent, incomplete, and hard to clean.
- Time-consuming analysis and manual reporting.
- Difficulty identifying the most important risk factors.
- Lack of interactive dashboards for quick insights.
- Hard to communicate medical findings to non-technical stakeholders.

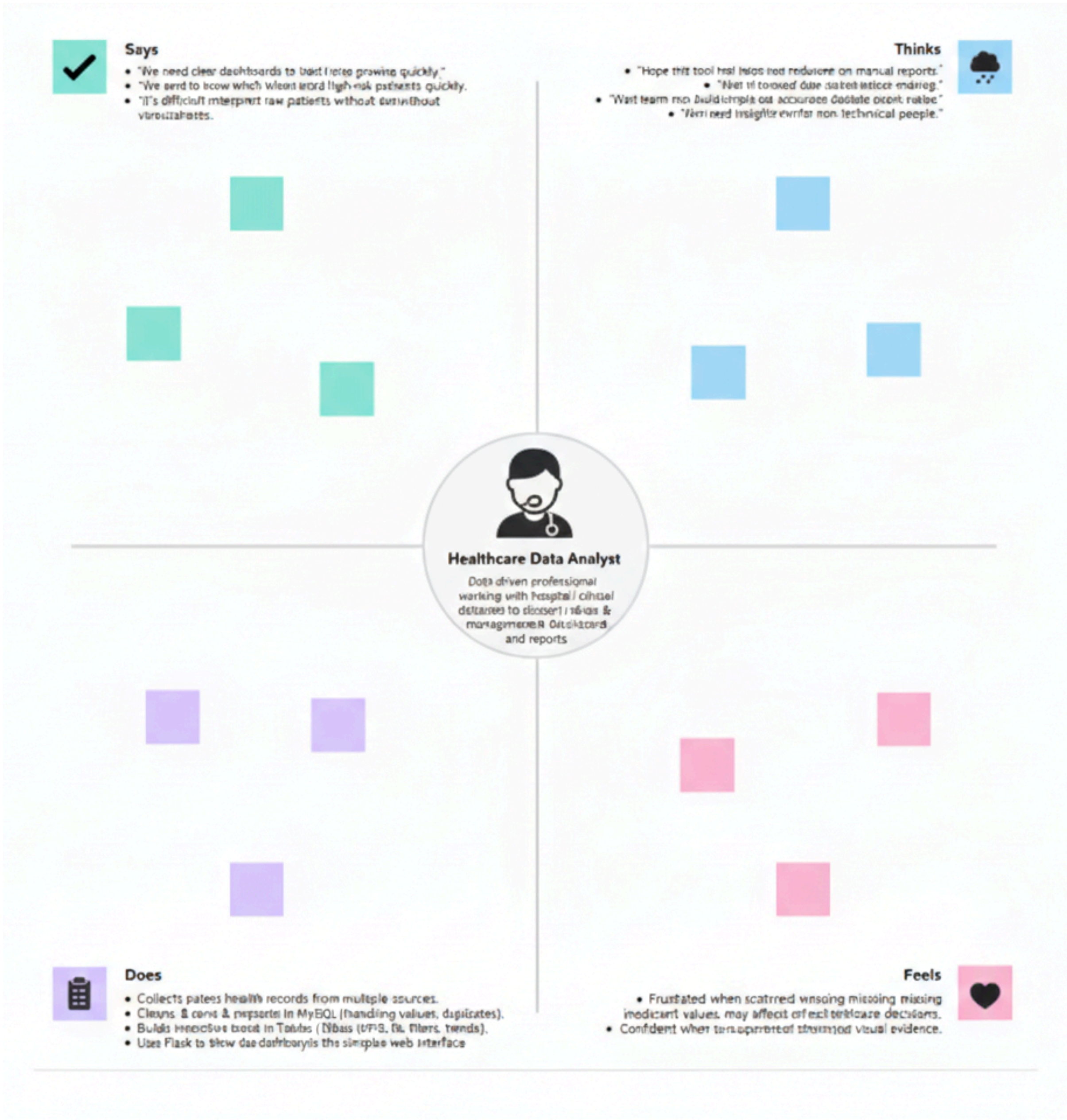
GAINS

- Interactive Tableau dashboards for easy risk factor analysis.
- Faster decision-making for doctors and hospital management.
- Clear understanding of high-risk patient groups.
- Better public health awareness through visual storytelling.
- Efficient reporting and reduced manual workload.
- Stronger healthcare strategies based on real patient data.

Solution Statement

“Heart Disease Insights” empowers healthcare professionals and analysts to explore, visualize, and understand heart disease risk factors using MySQL, Tableau, and Flask transforming scattered patient data into actionable insights for early detection, prevention, and smarter healthcare decisions.

Example:



2.3 Brainstorming

Brainstorm & Idea Prioritization:

During the brainstorming phase of the **Heart Disease Analysis** project, I explored multiple ideas to understand the key factors that influence heart disease risk and to help users identify major health patterns from the dataset.

I focused on important analysis areas such as **age-wise heart disease distribution**, **gender-wise comparison**, **impact of lifestyle habits (smoking, alcohol, physical activity)**, and **health condition factors (diabetes, BMI, general health)**. I also planned to use interactive visualizations such as **bar charts, line charts, heatmaps, and comparison dashboards** in Tableau to clearly show the relationship between risk factors and heart disease.

The main goal was to create an **interactive Tableau dashboard and story** that provides meaningful insights for users and supports better understanding of heart disease trends using data analytics.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

In the first step, the project was planned and initiated individually by analyzing the problem domain and identifying the most important factors related to heart disease. Since this is an individual project, all tasks such as **data preparation, SQL analysis, Tableau visualization, and Flask integration** were handled by me.

After reviewing the dataset and internship project requirements, I finalized the problem statement based on real-world relevance and the need to identify major heart disease risk factors.

problem statement:

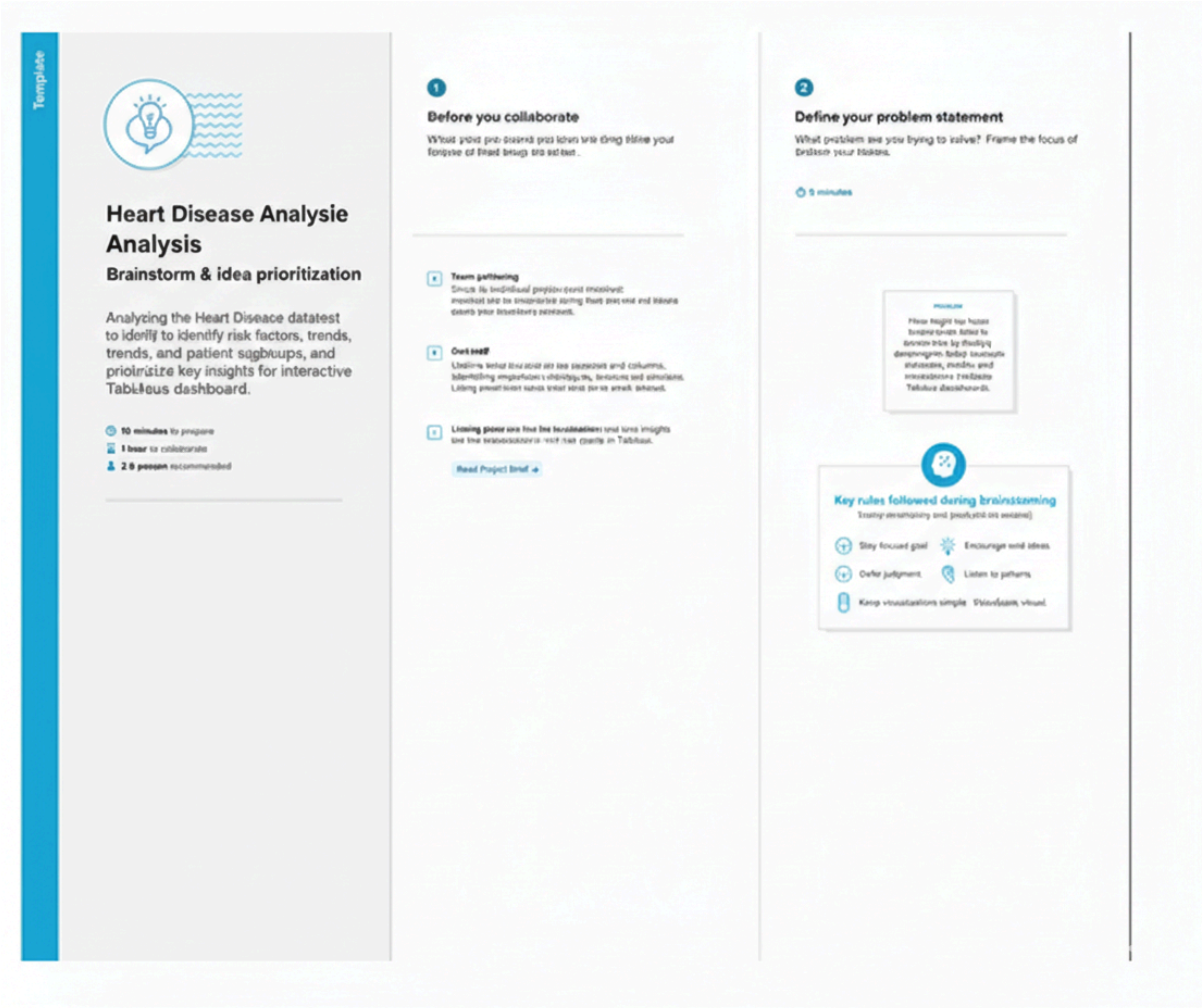
"Analyzing heart disease risk factors and trends using data visualization and analytics."

Team Leader : Shaik Mohammad Asif

Team member : Manasa G

Team member : Mavilla Nagamma

Team member : Mohan S

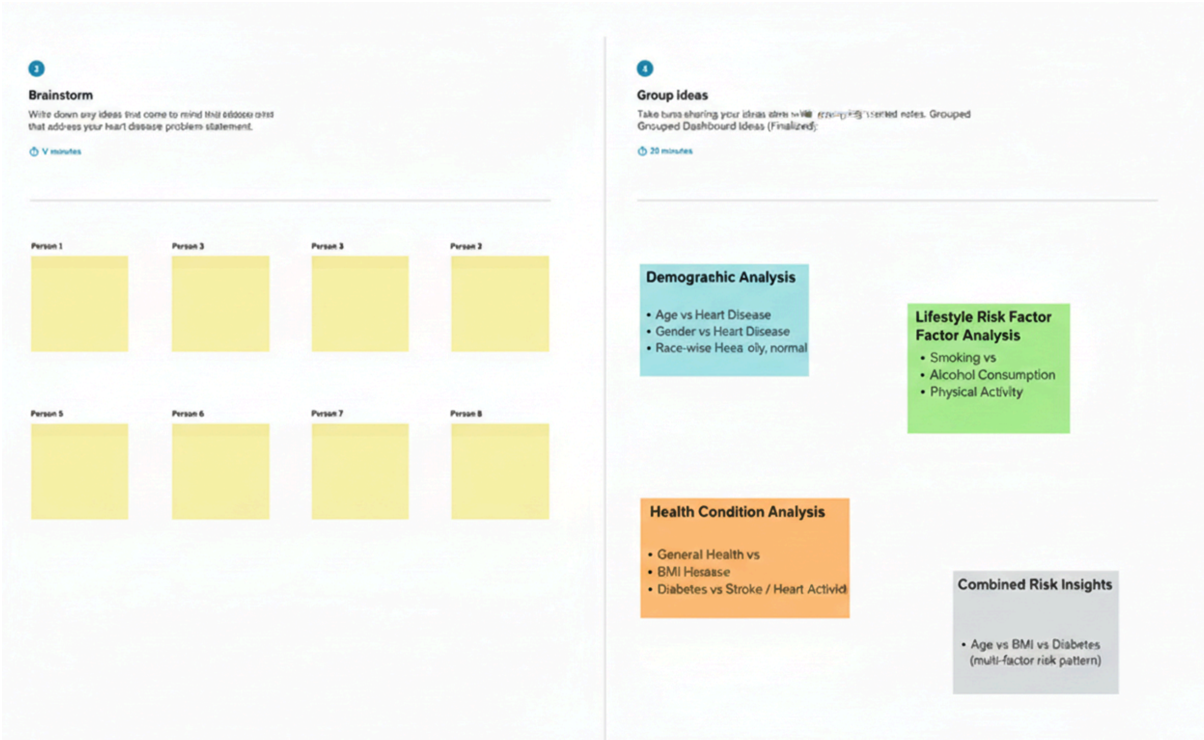


Step-2: Brainstorm, Idea Listing and Grouping

In this step, I listed multiple ideas related to analyzing the heart disease dataset to identify the most important patterns and risk factors. The initial ideas included **age-wise heart disease analysis, gender-wise comparison, BMI impact, diabetes relationship, lifestyle habits (smoking and alcohol), physical activity influence, general health condition, and race-wise heart disease distribution**.

After listing the ideas, I grouped them into meaningful categories such as **Demographic Factors, Lifestyle Factors, and Health Condition Factors**. This grouping helped me focus on the most valuable insights and decide the key features required for building an interactive Tableau dashboard and story.

Then, I finalized the grouped insights into dashboard features such as **filters for age and gender, comparison charts for smoking and alcohol, health-condition analysis (diabetes, BMI, general health), and race-wise heart disease trends** to support better understanding of heart disease risk factors.



Step-3: Idea Prioritization

In this step, I prioritized the analysis ideas based on **project goals, data availability, and the importance of insights for understanding heart disease risk factors**. From the shortlisted ideas, I selected the most impactful features that could clearly explain the relationship between demographic, lifestyle, and health-condition variables with heart disease.

The following key insights were prioritized for the dashboard and story:

1. **Age & Gender Insight :**

   I prioritized analyzing heart disease trends across different age groups and gender, to understand which groups show higher risk and how the disease pattern changes with age.

2. **Lifestyle Habits Impact :**

   I selected lifestyle-based analysis such as:

   - Smoking vs Heart Disease
   - Alcohol Consumption vs Heart Disease
   - Physical Activity vs Heart Disease

     This was prioritized because lifestyle factors strongly influence heart health and are important for real-world interpretation.

3. **Health Condition Relationship :**

   I prioritized health-related risk factors including:

   - Diabetes vs Heart Disease / Stroke
   - BMI vs Heart Disease
   - General Health vs Heart Disease

These features were selected because they help identify medical conditions that are closely linked with heart disease.

4. **Race-wise Heart Disease Analysis :**

   I included race-based analysis to understand how heart disease distribution varies across different racial groups and to highlight population-level patterns.

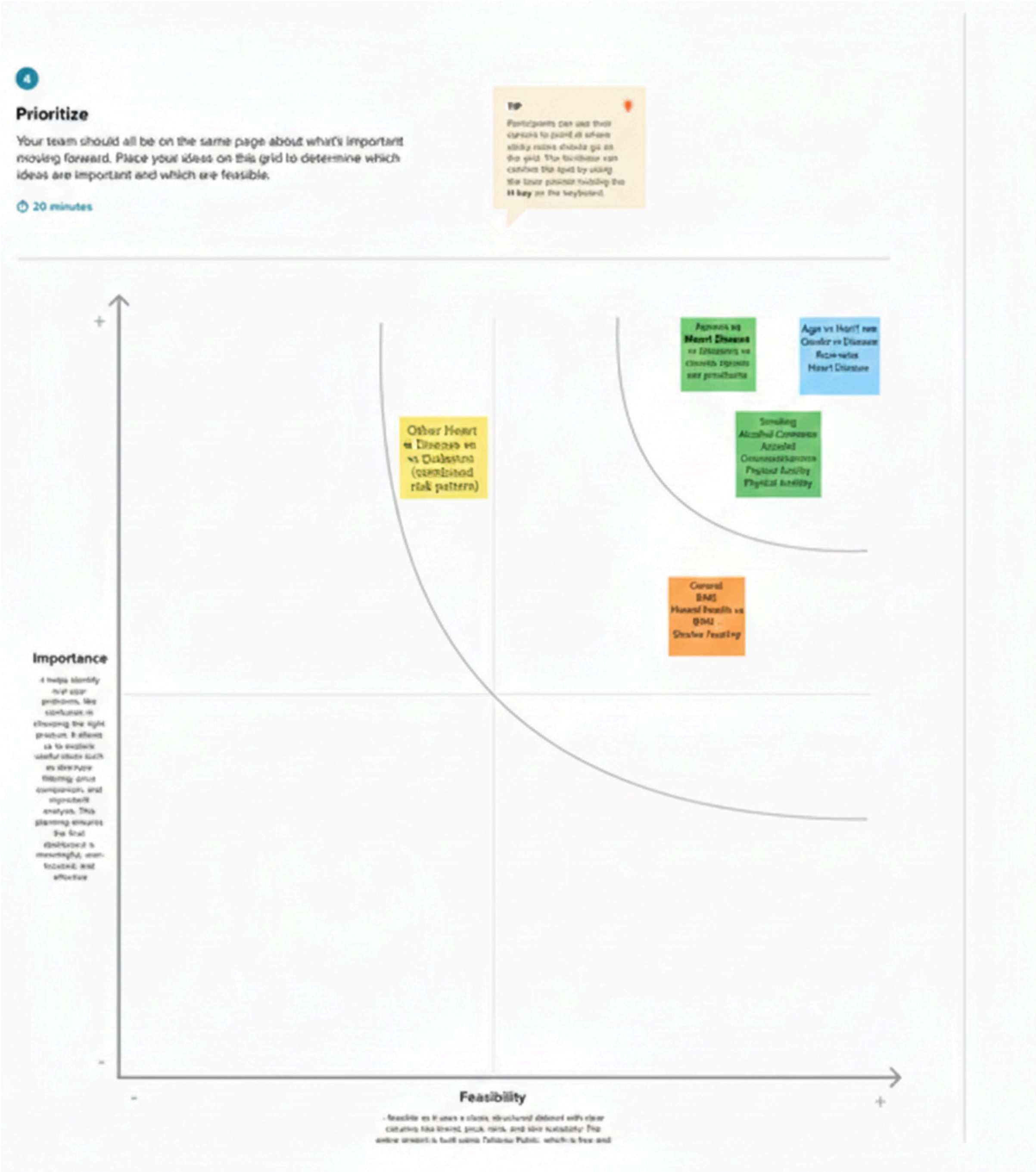
5. **Combined Risk Factor Insights :**

   Finally, I prioritized multi-factor analysis such as:

   - Age vs BMI vs Diabetes

     This was selected to show how multiple conditions together can increase heart disease risk.

These prioritized ideas helped me finalize the most meaningful dashboard insights and ensured the project stayed aligned with the main goal: **identifying and visualizing heart disease risk factors using Tableau and data analytics.**



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

3.2 Solution Requirement

Functional Requirements: Heart Disease Analysis

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Collect patient medical records dataset (age, gender, symptoms, vitals) Collect clinical test results (cholesterol, blood pressure, ECG, glucose, etc.) Collect target label (heart disease presence / absence)
 FR-2	Data Cleaning & Processing	Clean raw data (remove duplicates, handle missing values) Transform and aggregate data for analysis (age groups, risk categories)
FR-3	Data Storage	Store raw dataset securely in MySQL Store cleaned and processed dataset in structured tables
FR-4	Data Visualization & Analysis	Build interactive dashboards in Tableau

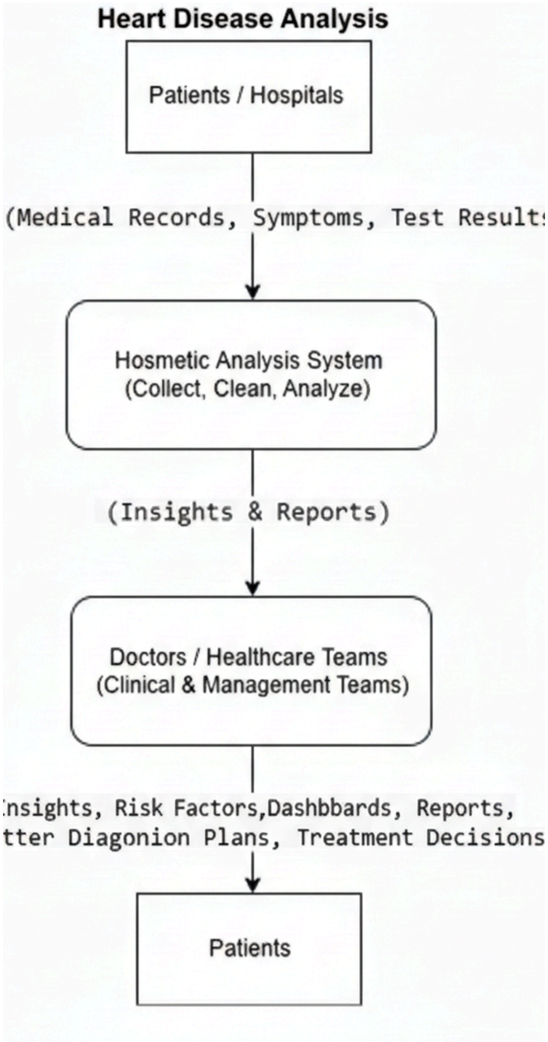
		Visualize risk factors (age, cholesterol, BP, diabetes, smoking, etc.) Provide downloadable reports / dashboard screenshots for documentation
FR-5	Alerts & Insights Delivery	Display dashboards and key insights through Flask web application Embed Tableau dashboard link inside Flask

Non-Functional Requirements: Heart Disease Analysis

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Tableau dashboards and Flask web pages should be easy to use and simple to understand for doctors and analysts.
NFR-2	**Security**	Patient health data must be stored securely in MySQL and should be accessed only by authorized users.
NFR-3	**Reliability**	The system should provide accurate and consistent results without dashboard errors or wrong insights.
NFR-4	**Performance**	Dashboards and visualizations should load quickly (within 5 seconds) for normal datasets.
NFR-5	**Availability**	The system should be available most of the time during working hours for analysis and reporting..
NFR-6	**Scalability**	The system should support adding more patient records and new medical factors in the future without issues.

3.3 Data Flow Diagram



1. Patients / Hospitals Provide Data

- **Data:** Patient medical records, symptoms, and test results (age, BP, cholesterol, ECG, diabetes, etc.).
- **How:**Collected through hospital records, health checkups, and clinical reports.

2. Heart Disease Analysis System (Process)

- **Actions:** Collects, cleans, stores, and analyzes patient data.
- **Tools:** Uses **MySQL** for data storage and cleaning, and Tableau for visualization and analysis.

3. Output: Insights & Reports

- **Outcome:** Generates interactive dashboards and reports showing heart disease risk patterns and major contributing factors.

4. Doctors / Healthcare Teams Use Insights

- **Teams:** Doctors, hospital management, healthcare analysts.
- **Actions:** Use insights to identify high-risk patients, support diagnosis, and plan prevention strategies..

5. Result: Better Healthcare Decisions

- **Outcome:** Patients receive improved screening, early detection, and better treatment or lifestyle recommendations.

6. Feedback Loop

- Doctors and hospitals update new patient data regularly, and the system continues improving with new records and analysis.

3.4 Technology Stack

Technical Architecture:

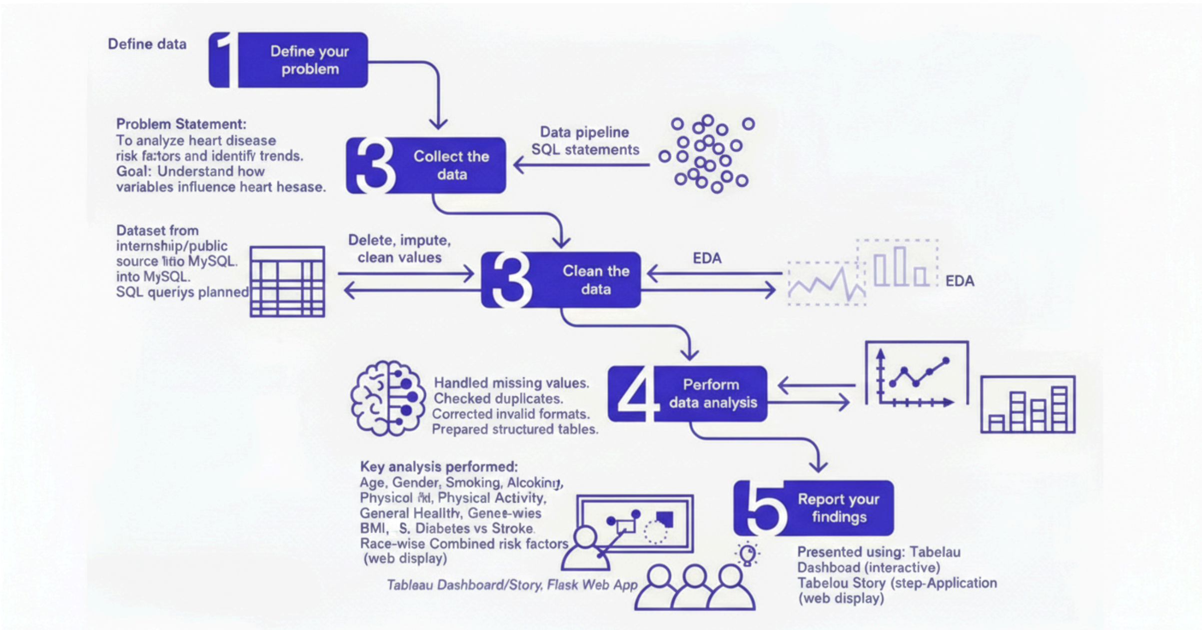


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for viewing dashboards and insights	HTML, CSS, JavaScript, Tableau Public Embedding
0.	Data Processing Logic	Data cleaning & preprocessing scripts	Python (Pandas, NumPy)
0.	Data Storage	Stores raw data and cleaned datasets	CSV files + MySQL Database

0.	Visualization Layer	Creates interactive visual dashboards and charts	Tableau Public / Tableau Desktop
0.	Infrastructure (Server / Hosting)	Hosts any scripts and serves embedded dashboards	Local Machine or Cloud Hosting (Render / Railway)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses open-source Python libraries for data processing	Python (Pandas, NumPy), Flask
2	Security	Data is stored securely in MySQL and dashboards shared via controlled links	MySQL authentication, Tableau Public permissions
3	Scalable Architecture	3-tier architecture: UI → Backend → Database	Flask + MySQL + Tableau
4	Availability	Dashboards accessible anytime through Tableau Public and Flask deployed link	Tableau Public, Render / Railway
5	Performance	Small dataset loads fast; Tableau extracts improve dashboard speed	Tableau Data Extracts, Python ETL

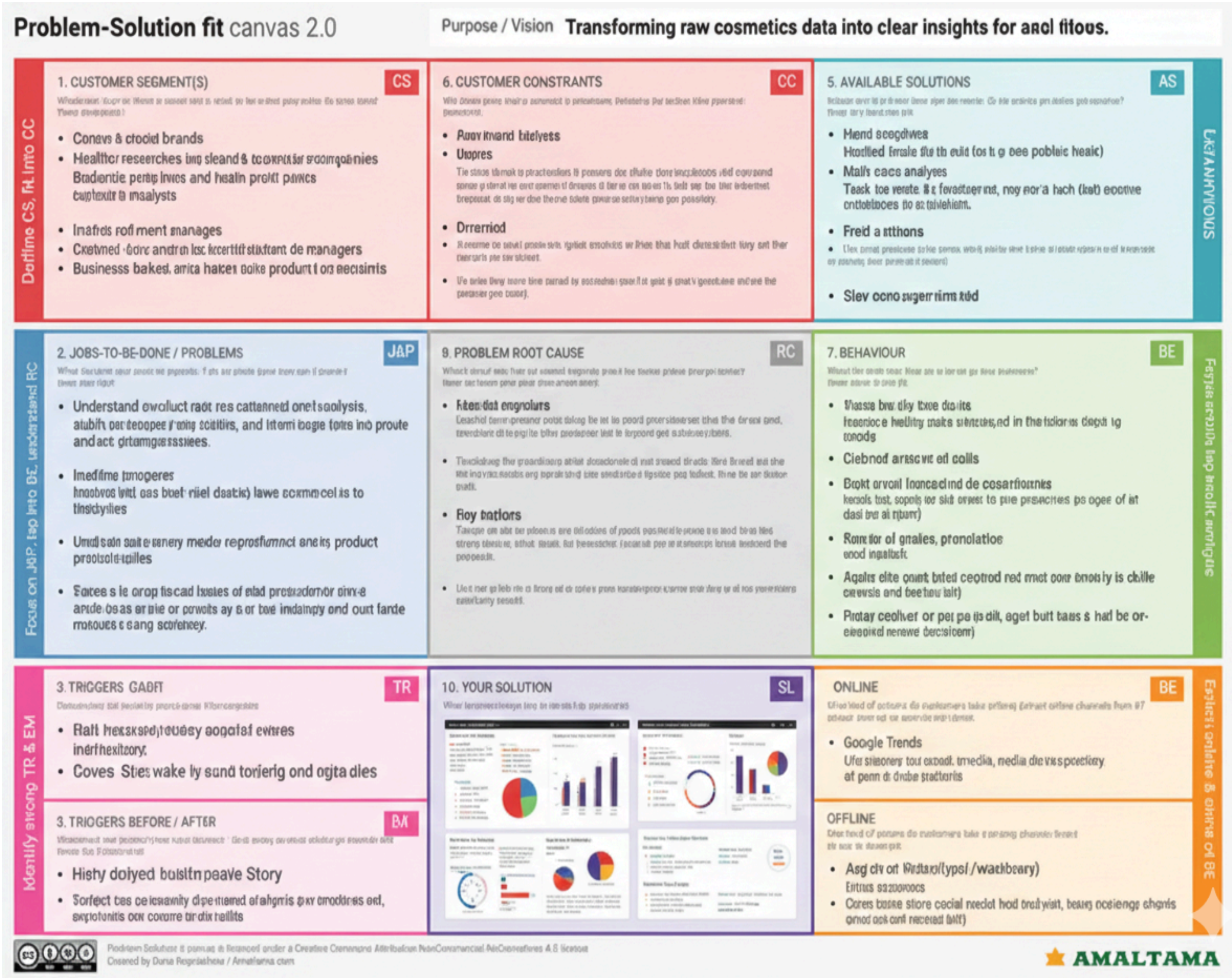
4. PROJECT DESIGN

4.1 Problem–Solution Fit

Purpose:
To solve the challenge faced by hospitals, healthcare providers, and patients in identifying heart disease risk factors early by providing interactive Tableau dashboards that deliver clear insights and support preventive healthcare decisions.

How it fits:

- Identifies real problems: heart disease risk depends on multiple factors and is hard to analyze manually.
- Uses existing behavior: hospitals already collect patient data (age, cholesterol, BP, ECG etc.).
- Fits user constraints: dashboard is simple to use, no advanced ML knowledge required.
- Helps faster decisions: doctors and analysts can quickly identify high-risk patterns.
- Supports prevention: improves awareness of risk factors such as chest pain type, cholesterol, age group, etc.


Problem-Solution fit canvas 2.0
Purpose / Vision
Transforming raw cosmetics data into clear insights for aacl fitous.
1. CUSTOMER SEGMENT(S)
CS
6. CUSTOMER CONSTRAINTS
CC
5. AVAILABLE SOLUTIONS
AS
Define CS, fit into CC
2. JOBS-TO-BE-DONE / PROBLEMS
J&P
9. PROBLEM ROOT CAUSE
RC
7. BEHAVIOUR
BE
3. TRIGGERS
TR
3. TRIGGERS BEFORE / AFTER
BA
10. YOUR SOLUTION
SL
ONLINE
BE
Google Trends
OFFLINE
Identify strong TR & EM
AMALTAMA

4.2 Proposed Solution

Proposed Solution :

S.No	parameter	Discription
1.	Problem Statement (Problem to be solved)	Heart disease is one of the leading causes of death worldwide. Many patients are unaware of their risk factors such as high cholesterol, blood pressure, chest pain type, ECG results, and age. Manual analysis is difficult, so there is a need for a dashboard-based solution to understand risk patterns and support early detection.
2.	Idea / Solution description	The solution uses Tableau dashboards to analyze heart disease patient data and visualize relationships between age, gender, cholesterol, blood pressure, chest pain, ECG, and heart disease presence. This helps identify risk factors quickly.
3.	Novelty / Uniqueness	This project combines medical attributes into interactive dashboards where users can filter and compare multiple risk factors dynamically, instead of relying on static charts or manual analysis.
4.	Social Impact / Customer Satisfaction	Helps increase awareness about heart disease risks, supports preventive healthcare, and enables early diagnosis. This can reduce severe heart conditions and improve patient outcomes.
5.	Business Model (Revenue Model)	Hospitals and health analytics companies can use the dashboard as a subscription-based analytics tool. It can also be offered as a clinical decision support dashboard for patient risk monitoring.
6.	Scalability of the Solution	The solution can be expanded to include more patient records, additional hospitals, and real-time streaming health data. It can also be extended to other diseases like diabetes or stroke risk analysis.

4.3 Solution Architecture

Solution Architecture:

The solution architecture of the Heart Disease Analysis project is designed to deliver meaningful insights using Tableau dashboards.

1. **Dataset Input :**

   The dataset contains patient information such as:

   - Age
   - Sex
   - Chest Pain Type (cp)
   - Resting Blood Pressure (trestbps)
   - Cholesterol (chol)
   - Fasting Blood Sugar (fbs)
   - Resting ECG (restecg)
   - Maximum Heart Rate (thalach)
   - Exercise-induced Angina (exang)
   - ST Depression (oldpeak)
   - Heart Disease Target (0/1)
2. **Data Cleaning & Preprocessing :**
   - Null values handled
   - Data types corrected
   - Duplicate records removed
   - Feature columns standardized
   - Exported cleaned dataset for Tableau
3. **Data Storage :**
   - Dataset stored in CSV
   - Imported into **MySQL database** for querying and analysis
4. **Dashboard Development (Tableau) :**

   Multiple worksheets created such as:

   - Age group vs heart disease
   - Gender distribution
   - Cholesterol vs heart disease
   - Chest pain type analysis
   - Blood pressure analysis
   - Max heart rate patterns
5. **Publishing & Sharing :**

   Dashboards and Story published in Tableau Public and embedded into Flask web app.

6. **End Users :**
   - Students / Analysts
   - Healthcare Researchers
   - Doctors (for understanding patterns)

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As an individual, I collected the heart disease dataset from reliable sources	2	High	Mohan
Sprint-1	Data Collection	USN-2	As an individual, I loaded and organized the collected dataset	1	High	Mohan
Sprint-1	Data Preprocessing	USN-3	As an individual, I cleaned missing values and removed duplicates in the dataset	3	High	Mohan
Sprint-1	Data Preprocessing	USN-4	As an individual, I handled categorical features and prepared the dataset for analysis	2	Medium	Mohan
Sprint-2		USN-5		5	High	Mohan

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
	Dashboard & Insights		As an individual, I built Tableau dashboards for heart disease and stroke analysis			
Sprint-2	Dashboard & Insights	USN-6	As an individual, I validated dashboard insights and tested filters	3	High	Mohan
Sprint-2	Deployment	USN-7	As an individual, I created HTML pages for embedding Tableau dashboards	3	Medium	Mohan
Sprint-2	Deployment	USN-8	As an individual, I deployed dashboards online using Flask	5	High	Mohan

Total Story Points:
Sprint-1: 8
Sprint-2: 16
Total: 24

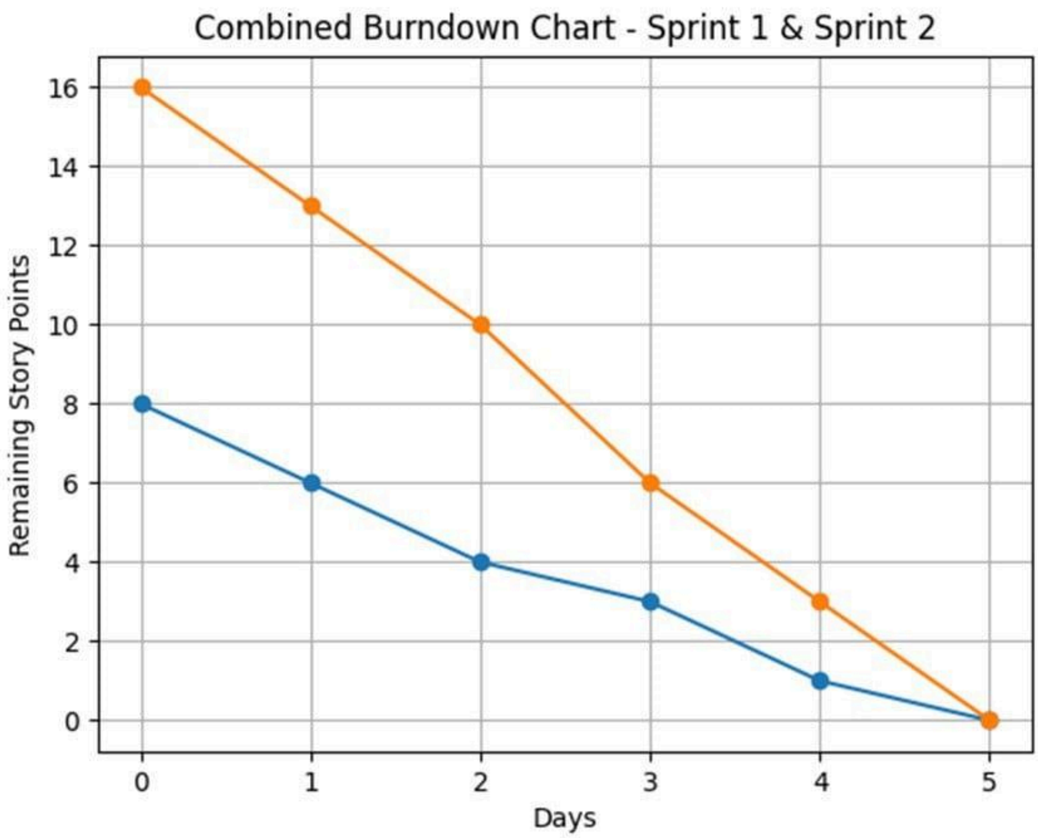
Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	sd	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	8 Days	10 Feb 2026	11 Feb 2026	8	15 Feb 2026
Sprint-2	16	8 Days	16 Feb 2026	20 Feb 2026	8	28 Feb 2026

Velocity:
Total Story Points = 24
Number of Sprints = 3
Velocity = 24 / 3 = 8 Story Points per Sprint

Average Velocity per Day:
Sprint Duration = 5 Days
Velocity per Day = 8 / 5 = 1.6 Story Points per Day

Burndown Chart:



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Model Performance Testing:

S.No.	Parameter	Values
1.	Data Rendered	Raw dataset includes Age, Sex, Chest Pain, BP, Cholesterol, ECG, Heart Rate, and Target column. ~300+ rows
0.	Data Preprocessing	Missing values handled, duplicates removed, categorical values encoded.
3.	Utilization of Filters	Filters applied: Gender, Age group, Chest pain type, Target, Cholesterol range.

4.	Calculation fields Used	Example: Heart disease % by age group, risk classification, average cholesterol by target.
5.	Dashboard design	**No of Visualizations / Graphs:** 9 **Dashboard 1**: Ranking Heart Disease Analysis (Activities 1.1, 1.3, 1.6, 1.8)
6	Story Design	**No of Visualizations / Graphs:** 9 1 dashboards combined into 1 story (Heart Disease Risk Factors + Overall Analysis)

Key Performance Metrics

Metric	**Description**
Dashboard Load Time	Time taken for Tableau Public dashboard to load
Visualization Rendering Time	Time taken to load charts
Filter Response Time	Time taken to update results after filters
Calculated Fields Evaluation	Time taken to compute formulas
Data Volume	Number of rows and columns processed

Test Results Summary

Test Scenario	**Observation**	**Status**
Dashboard Initial Load (Tableau Public)	3–5 seconds on average	✅ Pass
Filter Response (e.g., Gender = Female)	~1 second	✅ Pass
Story Scene Switch Time	2.3 seconds	✅ Pass
Visual Rendering with All Filters Applied	Slight lag on mobile, smooth on desktop	⚠️ Acceptable
Load on Flask Web Page	Fully rendered within 5–6 seconds (including embedded script)	✅ Pass

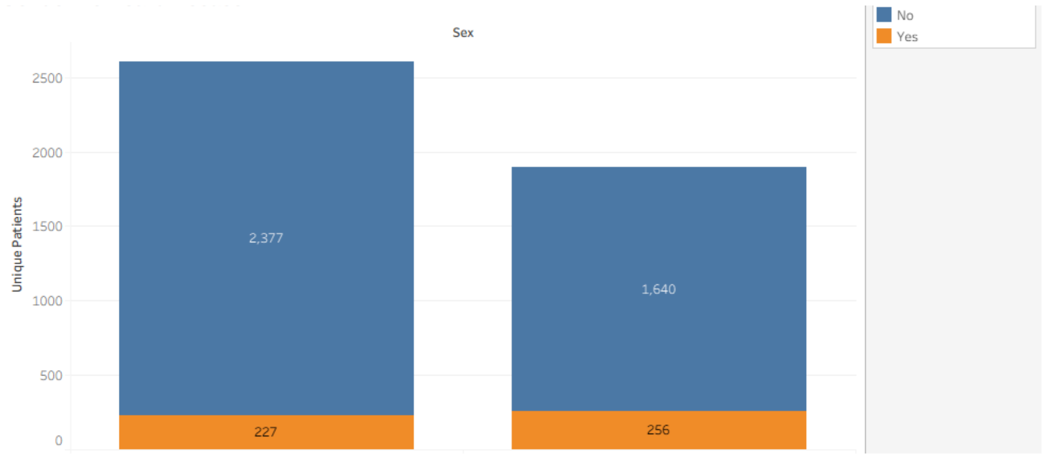
Recommendations for Optimization

Area	Optimization
Calculated Fields	Avoid complex LOD expressions
Filter Usage	Use extract filters to reduce scan
Dashboard Layout	Avoid overloading too many charts in one sheet
Data Volume Handling	Aggregate data before visualizing

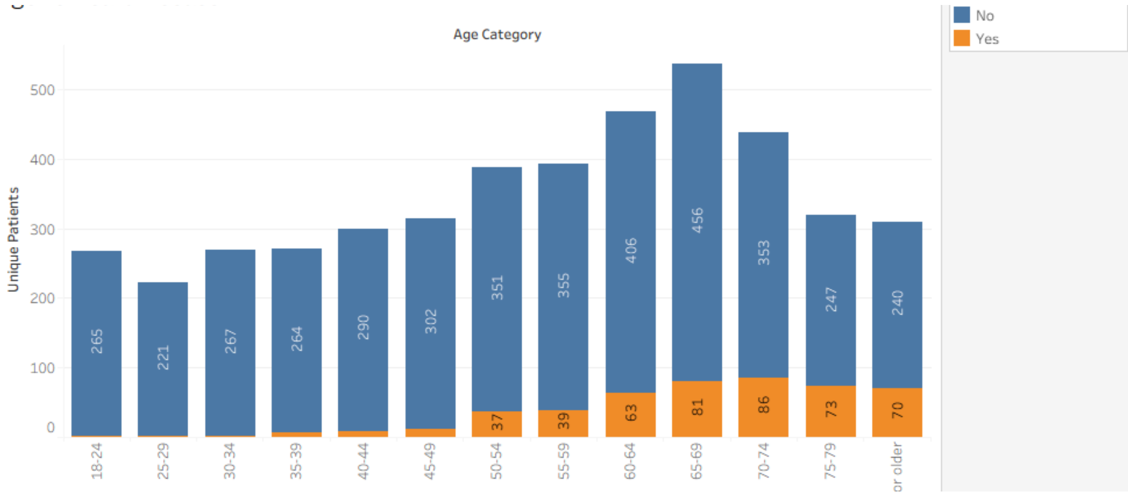
7. RESULTS

7.1 Output Screenshots

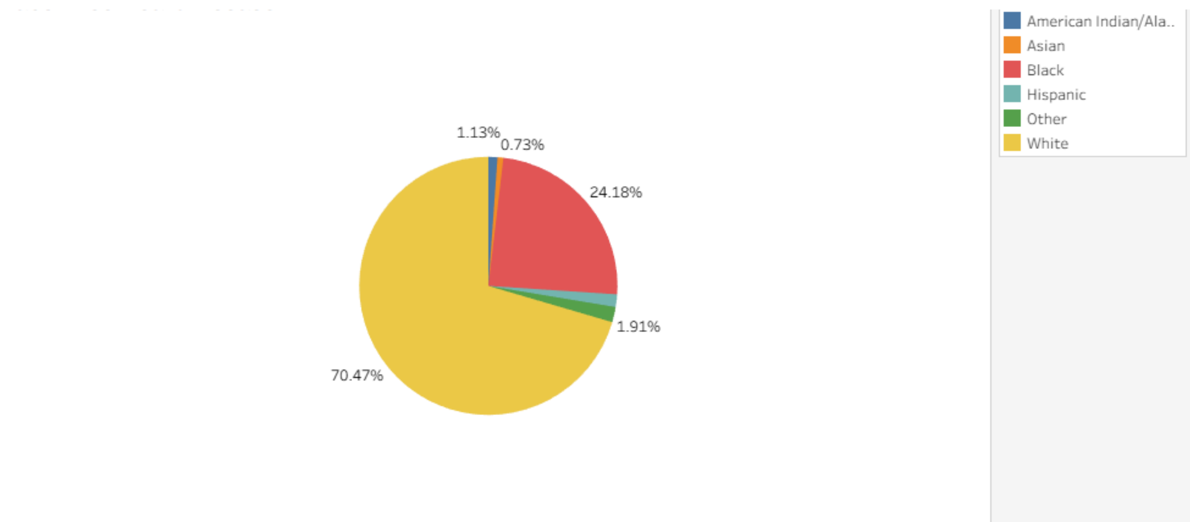
Activity 1.1: Age Group vs Heart Disease :



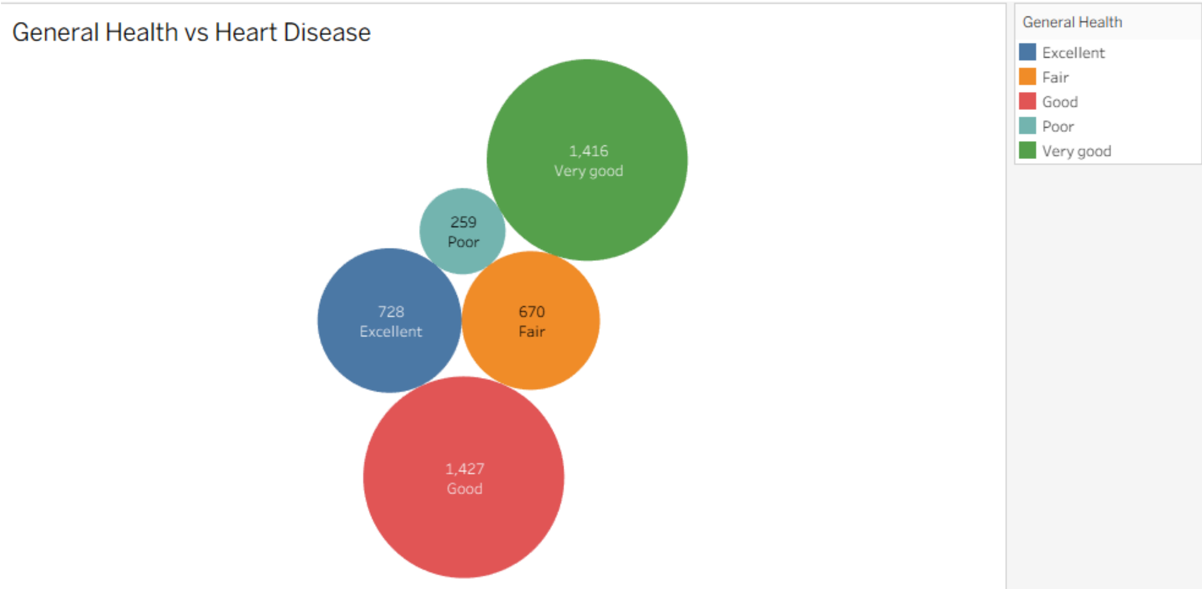
Activity 1.2: Age vs Heart Diseas



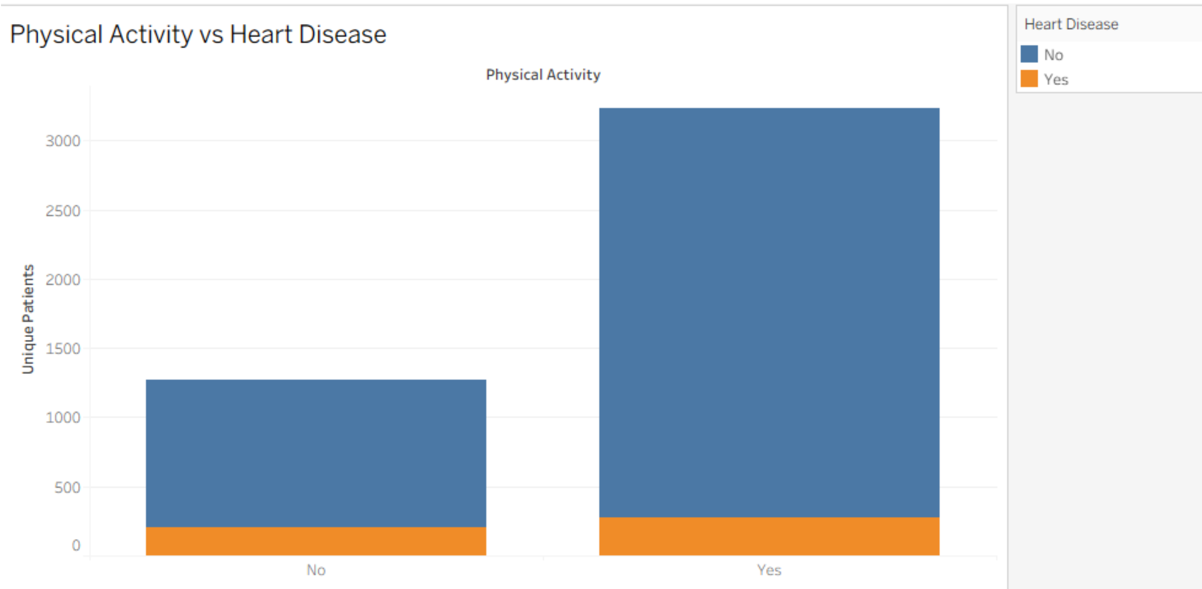
Activity 1.3: Race Wise Heart Disease



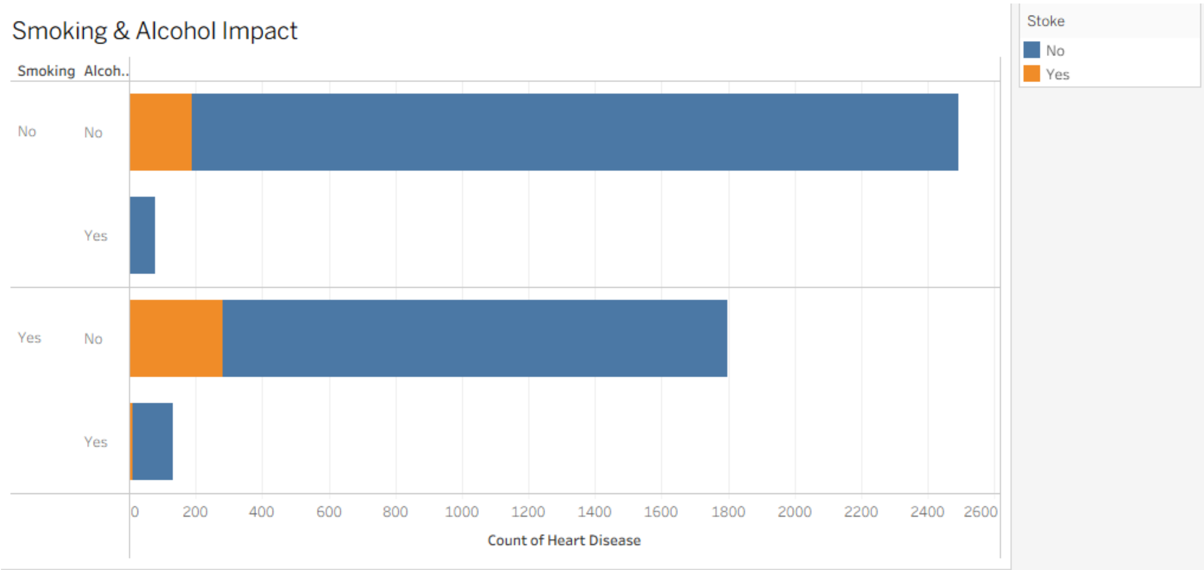
Activity 1.4 : General Health vs Heart Disease



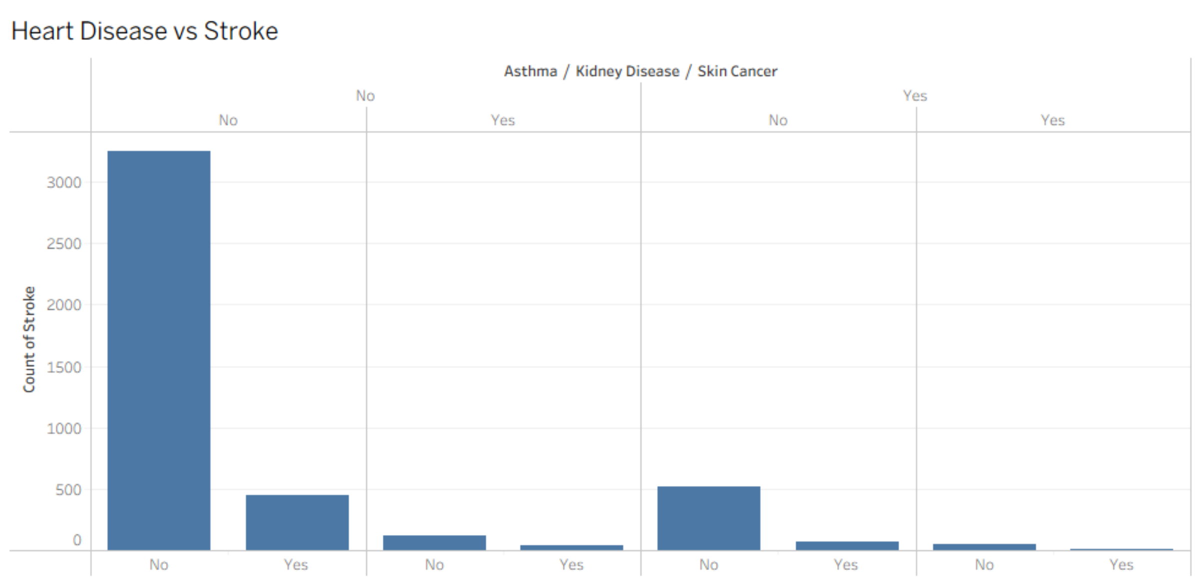
Activity 1.5 : Physical Activity vs Heart Disease



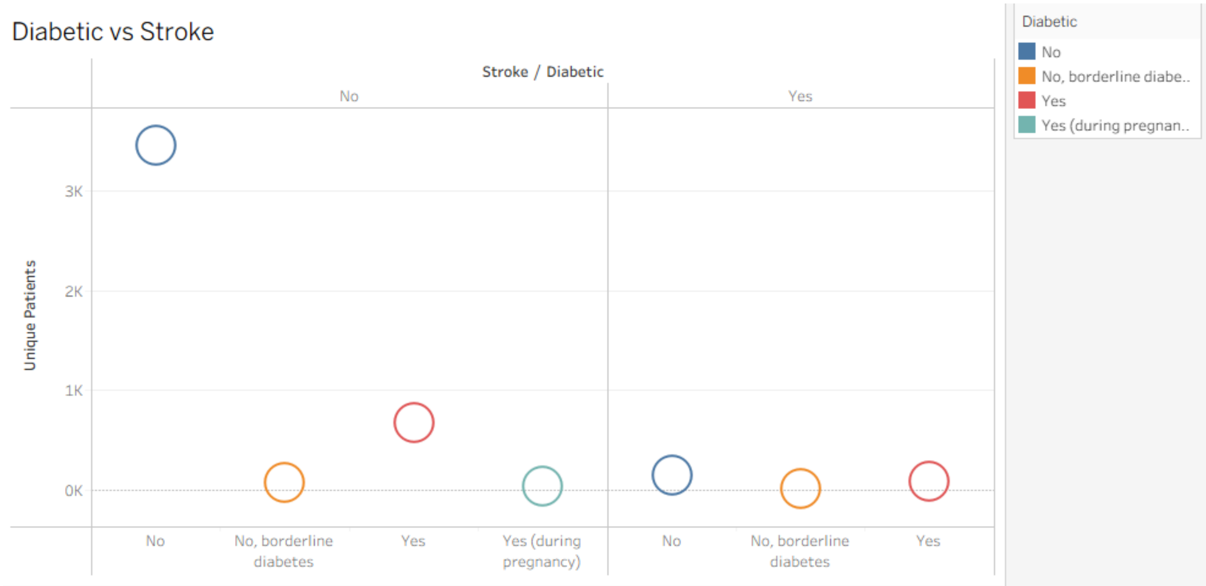
Activity 1.6 : Smoking & Alcohol Impact



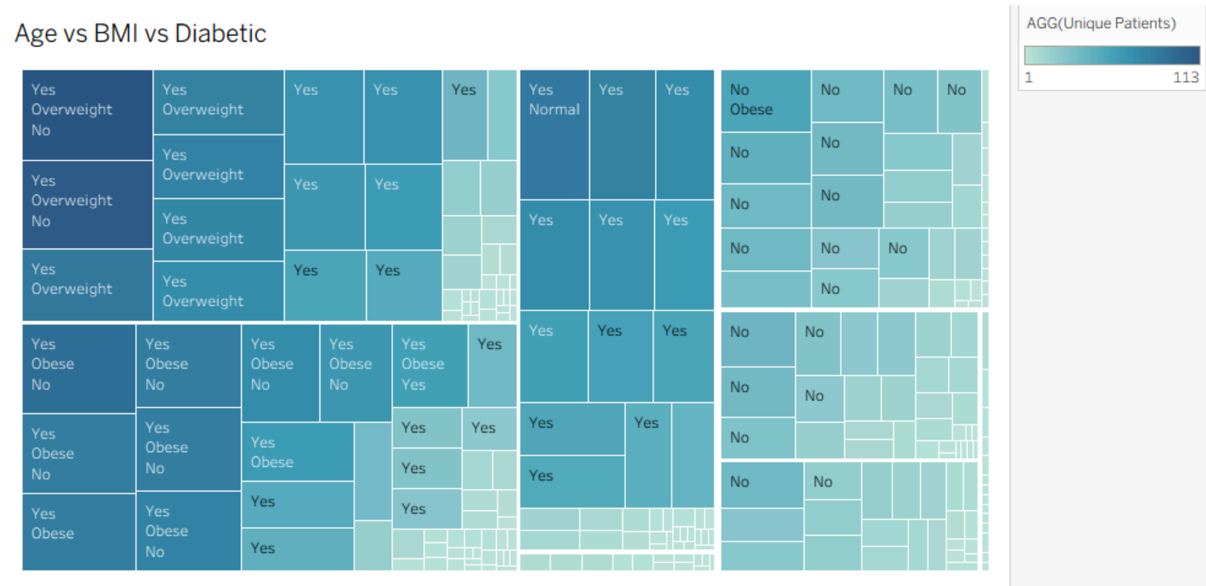
Activity 1.7 : Heart Disease vs Stroke



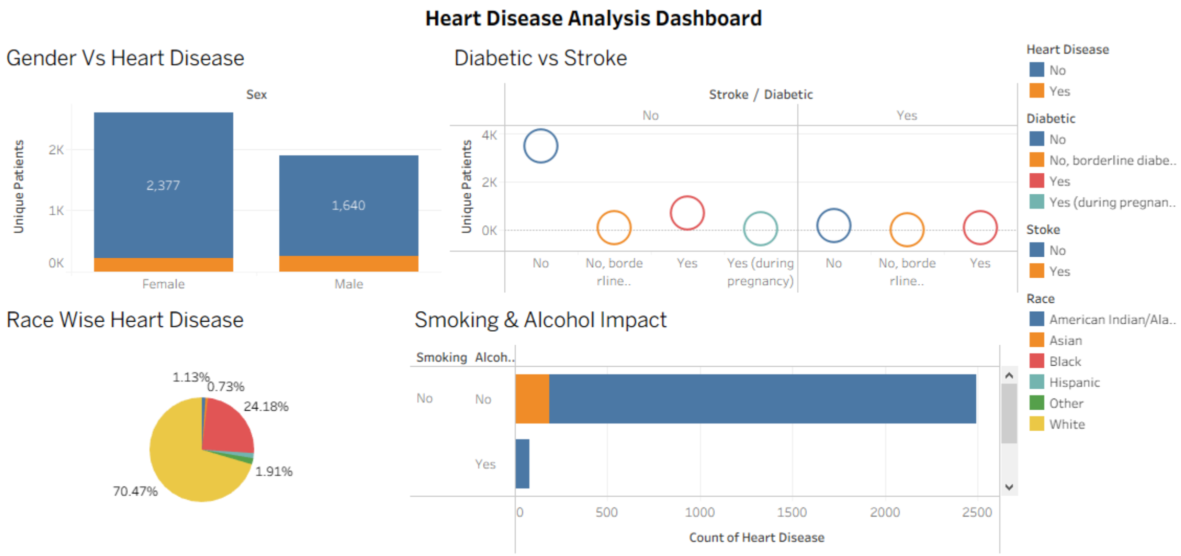
Activity 1.8 : Diabetic vs Stroke



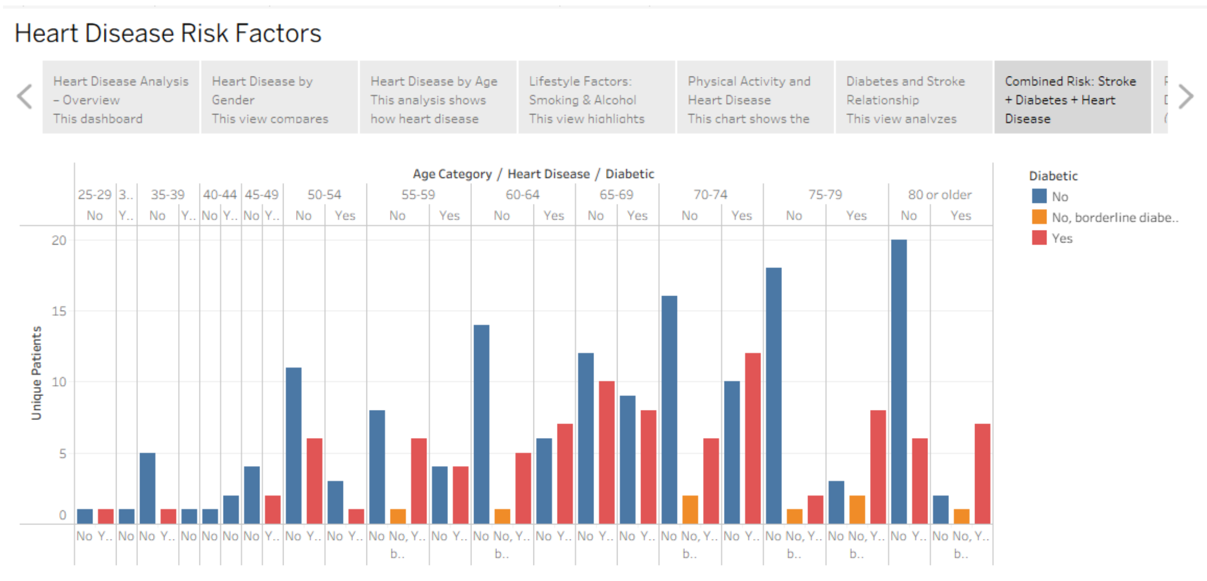
Activity 1.9 : Age vs BMI vs Diabetic



Ranking Heart Disease Analysis Dashboard :



Heart Disease Risk Factors Story :



8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Provides clear interactive visualizations for complex medical data.
- Helps understand key heart disease risk factors quickly.
- Enables filtering by gender, age, and clinical attributes.

- Easy to share dashboards using Tableau Public links.
- Supports data-driven preventive healthcare analysis.

8.2 Disadvantages

- Depends on dataset quality and accuracy.
- Tableau requires basic skills to modify dashboards.
- Limited medical accuracy (this is a student analysis project).
- Dataset size is small (demo-level).
- Does not include real-time hospital data integration.

9. CONCLUSION

The **Heart Disease Analysis** project successfully demonstrates how data visualization can help healthcare analysts and researchers identify important risk factors associated with heart disease. Using Tableau, the project transforms raw patient data into interactive dashboards and stories that highlight patterns related to age, gender, cholesterol, blood pressure, chest pain, and ECG results. These insights support better understanding, awareness, and early prevention of heart disease.

10. FUTURE SCOPE

- IIntegrate Machine Learning models to predict heart disease risk.
- Expand dataset size with real-world clinical data.
- Add real-time updates using APIs and hospital databases.
- Create a full web application with login access for hospitals.
- Extend the same dashboard approach to diabetes, stroke, and hypertension.

11. Deployment of Flask Web Application with Embedded Tableau Dashboard (Heart Disease Analysis)

11.1 Overview

This section explains the deployment process of the Flask-based web application developed for the **Heart Disease Analysis Project**.
The Flask application embeds an interactive **Tableau Public dashboard and Tableau story**, enabling users to explore heart disease insights through charts and filters.

The application was deployed using **Render.com**, a cloud hosting platform suitable for Python-based web applications.

11.2 Hosting Platform

- **Platform:** Render.com
- **URL:** https://render.com
- **Purpose:** To host the Flask application and make it available through a publicly accessible URL.
- **Reason for Selection:**

  Render was selected because it provides:

  - Free-tier hosting support
  - Native support for Python + Flask applications
  - Easy GitHub integration
  - Automatic deployment after code updates
  - Simple setup without complex DevOps configuration

11.3 Project Structure

The Flask application was organized in the following structure:

```
/flask
├── app.py              # Main Flask application logic
├── requirements.txt    # Project dependencies for deployment
├── Procfile            # Specifies how to run the app using Gunicorn
├── templates/
│   └── index.html      # HTML template embedding Tableau Dashboard and Story
├── static/             # Optional folder for CSS/JS or static assets
```

11.4 Key Configuration Files

11.4.1 requirements.txt

This file defines the Python dependencies required for the project.

Render uses this file to install the correct packages during deployment.

```
Flask==2.3.2
gunicorn==21.2.0
```

11.4.2 Procfile

This file instructs Render to run the Flask application using Gunicorn (production WSGI server).

```
web: gunicorn app:app
```

Note: `app:app` refers to:

- First `app` → filename (`app.py`)
- Second `app` → Flask instance inside the file

11.5 Deployment Process

The following steps were followed to deploy the Heart Disease Flask application:

1. **Repository Setup**
   The Flask project was uploaded to a public GitHub repository:
   🔗 github link for flaskapp
2. **Connecting to Render**
   - Logged into **Render.com** using GitHub credentials
   - Clicked **"New Web Service"**
   - **Selected the Flask GitHub repository and connected it to Render**
3. **Configuration Settings**
   The following deployment settings were configured in Render:
   - **Build Command:** pip install -r requirements.txt
   - **Start Command:** gunicorn app:app
   - **Runtime Environment:** Python 3 (auto-detected)
4. **Automatic Build & Deployment**
   After configuration, Render automatically:
   - Cloned the GitHub repository
   - Installed the required dependencies
   - Started the Flask application using Gunicorn
   - Generated a public URL to access the live Heart Disease Analysis web application

11.6 Issue Encountered and Resolution

During the initial deployment, the following error occurred:

```
ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'
```

Cause: The requirements.txt file was missing from the repository.

Resolution:

- Created the `requirements.txt` file manually
- Added Flask and Gunicorn dependencies
- Committed and pushed the file to GitHub
- Triggered a redeploy in Render

  After redeployment, the application ran successfully.

11.13 Final Result

After deployment, the Flask application successfully displayed the **embedded Tableau Public dashboard and Tableau story** for the Heart Disease Analysis project.

Dashboard URL: Heart Disease Dashboard (wait 2 mins loading takes time)

Story URL : Heart Disease Risk (wait 2 mins loading takes time)

11.8 Conclusion

The deployment demonstrates a simple and effective method to deliver interactive Tableau dashboards through a Flask web application hosted on Render.

This approach enables the Heart Disease Analysis insights to be accessed publicly through a lightweight and scalable solution, supporting real-time dashboard interaction without requiring local Tableau software.

All Links

Data set : Data Set Link

Tableau Viz Public Dashboard URL : Heart Disease Insights using tableau

Tableau Viz Public Story URL : Heart Disease Risk

GitHub link for flask : github link for flaskapp